

Assignment 3 — Instrumental Variables

Random Assignment as an Instrument: The “Hotspotting” RCT

Choose your method. For this assignment you complete **one** of three options — matching, instrumental variables, or difference-in-differences — whichever best fits the research you care about. All three share the structure below; this file is the **Instrumental Variables** option. Submit one knitted R Markdown file, acknowledge your group, and include the AI-use statement. **Four blocks:** A and D are your own thinking (most weight), B is AI-assisted execution, C tests whether you catch AI's mistakes. Late penalty 10%/day.

Video (★). The ★ part below requires a video of no more than 3 minutes explaining how you got your answer and what it means.

The data. Use *Dataset 3b_RCT* with outcome *post_ie_charges_180_IP* (180-day inpatient spending). The data contain the **randomization** variable but not the true treatment. Build a *treated* indicator from *link2care_duration* in *Dataset2b_RCT_AdditionalVariables* — count someone as treated if they participated at least 90 days. **Context:** the hotspotting program connects superutilizers to outpatient and social services; a 2020 RCT randomized enrollment, but participation was voluntary.

Research question. What is the effect of actually taking up the hotspotting program on 180-day inpatient spending?

Block A — Setup & Judgment (no code)

Choose a journal you like where a study like this could end up (clinical, econ, sociology, up to you). Using that journal's tone and structure, write the introduction to an interesting paper answering this question. You can limit yourself to 1-2 pages max here (no need to fill all that space either). Some possible questions you could address include:

- Why should anyone care?** In one paragraph, motivate this question from a physician's, a researcher's, and a policy-maker's vantage point. What would each do differently if they knew the answer?
- Anticipate the selection.** What is the underlying comparison in an ideal world, and what do you have in the data? Which confounders can you observe in the data, and which can you not? What are the main identification challenges?
- State the identifying assumption in one sentence.** Complete precisely: “A regression of spending on HDHP enrollment recovers the causal effect only if ...” Then give one reason to believe it and one reason to doubt it *in this setting*.

Block B — AI-Assisted Execution (use AI deliberately)

Use **AI to write the code** — fluent, well-directed use is the goal. For **one** part, paste in the prompt you gave, the raw code returned, and a sentence on what you corrected or redirected (your prompt + critique artifact).

- a. **Construct treatment and clean.** Build the *treated* variable (≥ 90 days) from *link2care_duration*. Choose covariates, measurement/transformations, and handle missing data; justify your choices.
- b. **Describe the groups.** Show a histogram of *link2care_duration* and a summary table across the three groups: control, randomized-but-not-treated, and randomized-and-treated. Decide which covariates you need and include them.
- c. **Naive regression.** Regress the outcome on *treated* directly, ignoring endogeneity. What does this estimate, and why is it likely biased?
- d. **IV / 2SLS.** Estimate the IV using randomization as the instrument for *takeup*. Report and interpret *both* the first stage (*takeup* on randomization) and the 2SLS estimate. How does it compare to the naive regression?

Block C — The Error Hunt (catch what AI gets wrong)

A student asked an AI assistant to run this analysis and pasted the reply below without checking it. **The code runs without any error message** and the numbers look plausible — but it has **at least three consequential problems** (a mix of coding bugs and interpretation mistakes).

Reproduce it on the real data, then find and fix the problems. For each: (i) what is wrong, (ii) the corrected code or statement, (iii) one sentence on how it changes the conclusion. Flag any extra problem you find that we did not plant.

The AI-generated analysis:

```
library(tidyverse); library(fixest)
df <- read_csv("Dataset3b_RCT.csv") # has `randomized`, `treated`, outcome

# First stage, then plug predicted takeup into the outcome equation
fs <- feols(treated ~ randomized, data = df)
df$treat_hat <- predict(fs)
ss <- feols(post_ie_charges_180_IP ~ treat_hat, data = df)
summary(ss) # default standard errors

# "The 2SLS estimate is the average effect of the program on inpatient
# spending for the whole superutilizer population."
```

Block D — Interpret & Defend

- a. **★ Translate for a decision-maker.** A program director asks: “If we get someone to actually enroll and stay, how much does their inpatient spending drop, and how sure are you?” Answer in plain language in your video. Be explicit about which patients your estimate speaks to (not necessarily everyone) and what that means for scaling the program.
- b. **Put your results into context.** Assume that you fully believe your results. Write the first two paragraphs of a Discussion section that interprets these results in a way that you think an editor would find compelling. You can compare them to external literature, report clinical/economic significance, or find some other “headline” way to showcase these results.

- c. **Revisit your assumption.** Return to your one-sentence identifying assumption. Which of the IV assumptions (relevance, exclusion, monotonicity) worries you most here, and why? Create and report at least one piece of evidence bearing on the assumption you can test (e.g., first-stage strength, or balance on randomization). Why is your evidence imperfect?

Mini Referee Report

Peer review is the part of the job no AI can do for you. AI assistants summarize happily but rarely land the concern that actually threatens a result — and they tend to agree with whatever they are reviewing. That judgment is yours.

Assigned paper: You may choose one paper from three options posted in our GitHub folder under HAD5744\Lecture08_IV2\RefereeReportOptions.

Read the assigned paper and write a short referee report (about one page, 400–600 words) naming the **two or three MAJOR concerns** — the ones that would change your confidence in the main result — not minor issues of writing. Structure it as: (1) a 2–3 sentence summary of the claim and how it is identified; (2) two or three major concerns, each with why it threatens the central result and what evidence would resolve it; and (3) a verdict naming the single most important revision.

Using AI here. You may use AI to understand the paper, but the concerns must be your own — its first list is usually generic and agreeable. If you used it, note in one line what it suggested and whether you agreed.